WHEN THE ROUTE BREAKS, WHAT NEXT?

SSCM GROUP 3 POSTER

Can dual sourcing keep the smartphone production line moving?

1. THE RESEARCH QUESTION

"How can dual sourcing improve resilience of the smartphone supply chain against Red Sea-related disruptions and what are the implications"

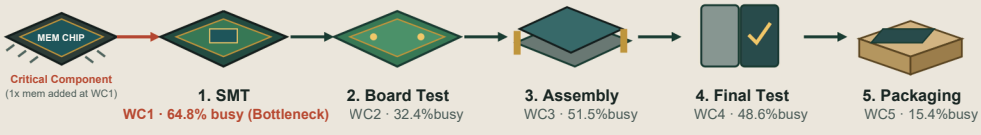
2. Product & Production Line in One Figure

Product: Samsung Galaxy S25 Ultra (256 GB) · Critical Component: Memory Chips (mem) · 52 Weeks (700 Smartphones/Week Target)



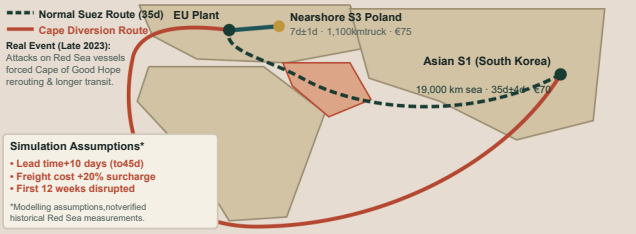
PRODUCT SPEC

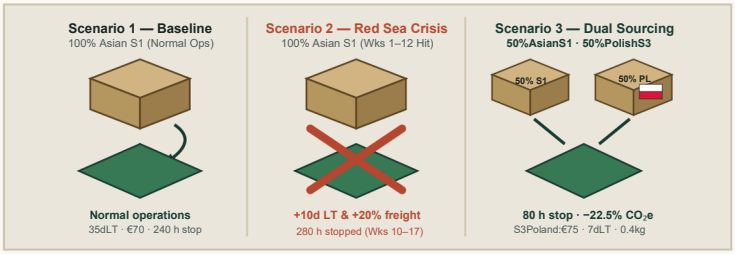
Samsung Galaxy S25 Ultra, 256 GB
Target: 619/wk (32,200/yr) 52 wks · 2,080 working hrs
Opening stock: 4,900 chips



3. The Disruption, the Real Event Behind It & Three Sourcing Scenarios

Late-2023 Red Sea Crisis Context, Model Assumptions (+10d LT, +20% Freight, Wks 1–12) & Countermeasure





4. The Scoreboard for All Three Scenarios & Cost Breakdown

Means & Lowest–Highest Ranges Across 10 Random Seeds (Seeds 1–10) Over 52 Weeks (Visual Centrepiece)

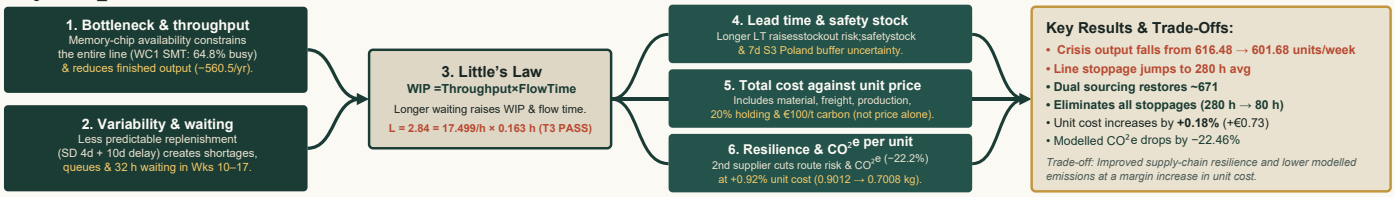
Performance indicator	Baseline	Red Sea Crisis	Crisis + Dual Sourcing
Units started	32,200	31,500	35,000
Units finished	32,196	31,598.0	34,998
Units per week	619.15	605.73	673.04
Total cost per unit	€402.13	€403.14	€402.83
CO_2e per unit	0.8904 kg	0.8904kg	0.6904 kg
Hours line stopped	240 h	280 h	80 h
Simulation runs	10	10	10
Minimum–maximum ranges (10 seeds)	**Baseline**	**Red Sea Crisis**	**Crisis + Dual Sourcing**
Units per week	605.73-619.19	592.25-605.73	666.29-673.04
Cost per unit	€402.07-402.46	€403.10-403.59	402.81-403.01
CO_2e per unit (kg)	0.8904-0.89040	0.8904-0.89040	0.6904-0.6904
Stopped hours (h)	240-280	280-320	80-100

Cost category (€)	Base	Crisis	Dual
Material	390.02	390.03	392.50
Freight	3.43	4.11	2.34
Production	8.44	8.64	7.75
Holding	0.29	0.51	0.29
Carbon	0.09	0.09	0.07

Course Values: Holding rate = 20%; Carbon price = €100/tonne CO_2e.
Why Dual Sourcing? Costs €0.73 (+0.18%) More:
→ Material costs increase by €2.50 per unit, while freight and production costs decrease. Holding and carbon costs also decrease slightly. Overall, total costs increase only marginally.
CO_2e Scope (Baseline → Dual):
CO_2e per unit: 0.8904 → 0.6904 kg
Result: 22.46% lower CO_2e emissions per smartphone.

5. Course Ideas Explaining Each Result & Key Implications

Bottleneck · Variability & Waiting · Little's Law · Lead Time & Safety Stock · Total Cost vs. Unit Price · Resilience & CO_2/Unit



Dual sourcing improved supply-chain resilience by restoring production output substantially reducing production stoppages. It also reduced modelled CO_2 emissions, although at a modest increase in total cost per unit. (Costantino & Pellegrino, 2010; Wang et al., 2026)

Limitations & Transparency

- The **+10-day lead time**, **+20% freight surcharge** and **12-week disruption duration** are simulation assumptions, not verified historical measurements.
- The CO_2 calculation may **double-count transport emissions** because supplier emission factors in sources.csv may already include transport.
- Results are **model outputs** (weekly call-offs assumed), not actual Samsung operational data.

Academic References in APA-style

1. Costantino, N., & Pellegrino, R. (2010). Choosing between single and multiple sourcing based on supplier default risk: A real options approach. *Journal of Purchasing and Supply Management*, 16(1), 27–40. https://doi.org/10.1016/j.pursup.2009.08.001
2. Wang, X., Tang, J., Chen, B., & Hou, B. (2026). Budget-constrained dual sourcing with option contracts under disruption risk. *Omega*. https://doi.org/10.1016/j.omega.2026.103682